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Relativistic Kramers-Unrestricted Exact-Two-Component Density Matrix Renormalization Group

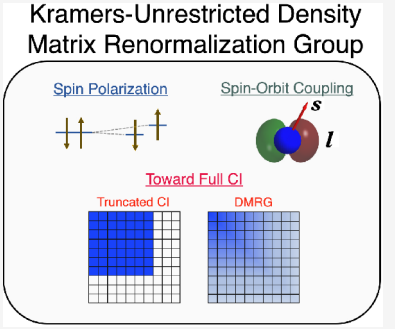
Chad E. Hoyer, Hang Hu, Lixin Lu, Stefan Knecht,* and Xiaosong Li*

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ABSTRACT: In this work we develop a variational relativistic density matrix renormalization group (DMRG) approach within the exact-two-component (X2C) framework (X2C-DMRG), using spinor orbitals optimized with the two-component relativistic complete active space self-consistent field. We investigate fine-structure splittings of p- (Ga, In, Tl) and d-block (Sc, Y, La) atoms and excitation energies of monohydride molecules (GeH, SnH, and TlH) with X2C-DMRG calculations using an all-electron relativistic Hamiltonian in a Kramers-unrestricted basis. We find that X2C-DMRG yields accurate ^{2}P and ^{2}D splittings compared to multireference configuration interaction with singles and doubles (MRCISD). We also investigated the degree of symmetry breaking in the atomic multiplets and convergence of electron correlation in the total energies. Symmetry breaking can be large in some cases (∼30 meV); however, increasing the number of renormalized block states m for the DMRG optimization recovers the symmetry breaking by several orders of magnitude. Encouragingly, we find the convergence of electron correlation to be close to MRCISDTQ5 quality. Relativistic X2C-DMRG approaches are important for cases where spin−orbit coupling is significant and the underlying reference wave function requires a large determinantal space. We are able to obtain quantitatively correct fine-structure splittings for systems up to 10^{19} number of determinants with traditional CI approaches, which are currently unfeasible to converge for the field.



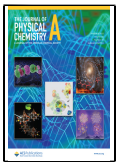
1. INTRODUCTION

Treatment of both electron correlation and spin−orbit coupling (SOC) is an active area of research because of applications in spectroscopy and heavy-element chemistry, from organic photochemistry to treatments of molecules containing late-row atoms.[1−4] Because degenerate or near-degenerate electronic states or orbitals are common in these applications, multireference approaches are needed as a starting point for treating electron correlation.[5] A common choice of multireference wave function is complete active space self-consistent field (CASSCF), where the number of configurations is determined by defining a space of electrons and orbitals (active space).[6] The remainder of the electron correlation (dynamic) can be recovered by post-CASSCF methods, such as CAS second-order perturbation theory (CASPT2).[7−9] The underlying exponential scaling in CAS methods motivates configuration interaction (CI) approaches with polynomial scaling,[10] such as density matrix renormalization group (DMRG).[11−15]

There are predominantly two approaches for all-electron treatments of spin−orbit coupling. In the first approach, the wave function is first variationally determined with no SOC, and then SOC is introduced via state interaction to couple selected states of interest.[16] The state interaction approach is usually used with the first-order SOC perturbation without additional orbital optimizations. Though very successful, the accuracy of the perturbative approach should decrease as the magnitude of SOC increases, which seems to occur with atoms below the fourth row.[17] Alternatively, SOC can be included variationally, which is commonly achieved by solving the Dirac equation.[18,19] As high orders of perturbations are included, orbitals are optimized with respect to SOC or the expansion space increases toward the full CI limit, and the perturbative approach should converge to the variational limit. Benchmarking of variational and perturbative treatments of SOC is an area of ongoing work.[20]

Development of variational relativistic multireference methods is an active area of research.[9,21−35] Although DMRG has found use in relativistic (four-component) Kramers-restricted calculations,[29,30] this is the first relativistic Kramers-unrestricted DMRG study. The Kramers-unrestricted and -restricted formalisms have their own advantages and disadvantages, depending on the subject matter. The advantage of the Kramers-restricted approach is that the Kramers' symmetry can be preserved with a constraint on the time-reversal symmetry throughout the calculation, but it cannot model cases where the Kramers' symmetry is physically broken (e.g., magnetic field[36−40]). The Kramers-unrestricted formal-

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ism, however, can model systems with broken time-reversal symmetry, but in the case where the Kramers symmetry persists it needs high-order excitation operators (e.g., in CAS) to recover the Kramers' symmetry.[41] In the relativistic regime where there are many close-to-degenerate minima, the Kramers-unrestricted formalism is able to explore a bigger parameter space and may give lower physical solutions.

In this work, we make use of the one-electron exact-two-component transformation (X2C) in our treatment of the Dirac equation and use the Boettger factor[42] to take into account two-electron spin−orbital effects.[32,33,43−62] Variational relativistic X2C-CASSCF[31] and X2C-MRCI[32] approaches are well established in the literature, and we relegate their description here to brief reviews of the X2C transformation in the Appendix. In the following section, we focus on the generalization of the DMRG approach to the Kramers-unrestricted relativistic two-component framework.

2. METHOD

The target multiconfigurational wave function $|\Psi\rangle$ obtained within a DMRG approach is a linear combination of configurations where the underlying molecular spinor orbitals are variationally optimized in the two-component relativistic framework,

$$|\Psi\rangle = \sum_{I} C^{I_1 I_2 \cdots I_N} |\mathbf{I}\rangle,\ |\mathbf{I}\rangle = |I_1, \ldots, I_N\rangle \tag{1}$$

Here, $C^{I_1 I_2 \cdots I_N}$ are complex-valued configuration-interaction coefficients, N is the number of active orbitals, and $|\mathbf{I}\rangle$ is an occupation number vector composed of Kramers-unrestricted molecular spinor orbitals (cf. eq 5). Because we are working in a Kramers-unrestricted spinor basis,[41] the occupation for an orbital j is limited to occupied (occ) or unoccupied (vac),

$$I_j \in \{|\mathrm{occ}\rangle, |\mathrm{vac}\rangle\} \tag{2}$$

as opposed to the often considered occupations, $|\uparrow\downarrow\rangle$, $|\uparrow\rangle$, $|\downarrow\rangle$, $|\mathrm{vac}\rangle$. As a result, the dimension of the coefficient tensor $\mathbf{C}$ is 2^N. Once the Kramers-unrestricted occupation scheme is defined, the existing complex-valued DMRG optimization algorithm can be used without further modification.

In a matrix-product-state (MPS) representation of $|\Psi\rangle$, we encode the CI coefficients $C^{I_1 I_2 \cdots I_N}$ as a product of $a_{j-1} \times a_j$-dimensional matrices $M^{I_j} = \{M^{I_j}_{a_{j-1} a_j}\}$, i.e., two matrices (corresponding to the local space dimension as indicated in eq 2) for the jth orbital,

$$\begin{aligned} |\Psi\rangle &= \sum_{I_1,\ldots,I_N} \sum_{a_1,\ldots,a_{N-1}} M^{I_1}_{1 a_1} M^{I_2}_{a_1 a_2} \cdots M^{I_N}_{a_{N-1} 1} |I_1, \ldots, I_N\rangle \\ &= \sum_{I} M^{I_1} M^{I_2} \cdots M^{I_N} |\mathbf{I}\rangle \end{aligned} \tag{3}$$

The last equality in eq 3 follows from collapsing the summation over the a_j indices (sometimes referred to as *virtual indices* or *bonds*) as matrix−matrix multiplications. Because the final contraction of the matrices M^{I_j} must yield a scalar coefficient $C^{I_1 I_2 \cdots I_N}$, the first and the last matrices are in practice $1 \times a_1$-dimensional row and $a_{N-1} \times 1$-dimensional column vectors, respectively. Allowing for the introduction of some maximum dimension m for the matrices M^{I_j}, with m commonly referred to as *number of renormalized block states*,[63] is the central idea that facilitates a reduction of the exponentially scaling FCI *ansatz* in eq 1 polynomially scaling MPS wave function *ansatz*.

As the dimension of m increases, the ansatz in eq 3 approaches the exact representation, where the innermost matrix has the maximum dimension of $m_{\max} = 2^{N/2}$ for N number of active orbitals in a Kramers-unrestricted basis. Note that this value of $m_{\max}$ holds for any CI problem with a variable number of electrons and does not represent the maximum m for a CI of fixed electron number; however, the authors are unaware of more practical ways of expressing the exact DMRG limit without resorting to any additional calculations. Notably, for quantum chemical problems, the dimension of M does not need to be close to $m_{\max}$ to achieve acceptable accuracy.[64−66]

By adjusting the upper limit of m in subsequent DMRG optimizations, we modify the degree to which the electrons are correlated. In the limit of converged m, DMRG reproduces CASCI, and if N is the total number of orbitals, DMRG reproduces full CI within numerical thresholds. For intermediate values of m, DMRG is lower bounded by CASCI or full CI because it is variational.[67]

Optimizations of excited states in quantum-chemical DMRG are not as straightforward as for traditional CI approaches.[68−73] In this work, we make use of a state-specific treatment (not to be confused with the treatment of orbitals in the state-average CASSCF optimization), which enforces electronic-state orthogonalization before and after each (two-site) tensor optimization for a given electronic state. For more information on the DMRG algorithms used in this work, see refs 30 and 74−76.

3. RESULTS AND DISCUSSION

The X2C-DMRG method is developed via a modular interface between the QCMaquis[74−76] and Chronus Quantum[77] computational packages. X2C-DMRG optimizations were performed in QCMaquis using the canonical spinor orbital basis optimized with the state-average X2C-CASSCF procedure in Chronus Quantum. In the X2C-DMRG optimization, spinor orbitals are ordered by energy; that is, we attempted neither to optimize the orbital ordering nor to select the spinor orbital spaces making use of quantum-information-theory based quantities. For comparison, exact-two-component multireference configuration interaction (X2C-MRCI)[32] calculations were also carried out.

The convergence criteria for X2C-CASSCF are 1×10^{-8} au in energy and 1×10^{-4} au in gradient, and the convergence criterion for traditional CI calculations is 1×10^{-6} in CI vector residue norms. For X2C-DMRG calculations, the convergence criterion for each state was 1×10^{-7} au.

We test our relativistic X2C-DMRG methodology on the fine-structure splitting of select p-block atoms (Ga, In, and Tl) and d-block atoms (Sc, Y, and La), previously shown to yield a robust test suite for relativistic CI methods[32] and p-block monohydrides (GeH, SnH, and TlH). The X2C-TZPAll-2c all-electron basis set was used for all calculations in this work.[78] For all p-block atom calculations, we used a X2C-SA(6)-CAS(3,8)SCF reference wave function, where SA(6) denotes averaging over six electronic states and the CAS(3,8) spaces consists of three electrons distributed in the eight valence s−p spinor orbitals. For d-block atoms, a SA(10)-CAS(3,12)SCF reference wave function was used, where the 12 orbitals correspond to the valence s−d manifold. For TlH, a SA(7)-CAS(4,10)SCF reference wave function was used, consisting of

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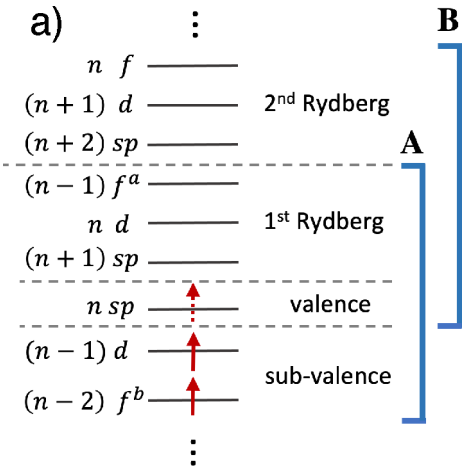
orbitals derived from the valence sp manifold of Tl and the 1s of H. The enlarged correlation space of TlH used for X2C-DMRG and X2C-MRCI was a (14,30) space, where correlating orbitals to the CAS(4,10) space and core Tl d orbitals were added. For the X2C-MRCISD calculation, RAS1 included the core Tl d orbitals, and RAS3 included the correlating orbitals to the (4,10). The GeH and SnH calculations used a SA(6)-CAS(1,6)SCF reference wave function derived from the p orbitals on the Ge/Sn. The X2C-DMRG and X2C-MRCI calculations used a larger correlation space (3,16), including the valence σ and σ^* orbitals and orbitals derived from correlating the p manifold of Ge/Sn. For X2C-MRCISD of GeH and SnH, the σ was added to RAS1, and the σ^* and correlating p orbitals were added to RAS3. All X2C-MRCISD calculations in this work constrained the maximum number of holes/electrons in RAS1/RAS3 to 2.

Multireference methods are needed for atomic fine-structure calculations, because all degenerate microstates are included in the relativistic wave function optimization. For atomic fine-structure calculations, we compare results computed using X2C-MRCI, X2C-CASCI, and X2C-DMRG in this work with the active spaces defined in Figure 1a and Figure 2a. The active spaces used in this work are well-defined schemes of introducing electron correlation to help us understand electron correlation recovery with DMRG. In addition to the active space, we also report the number of Slater determinants in the MRCISD (with the RAS1/RAS2/RAS3 partitioning reported in ref 32), denoted N^{MRCISD} in Figure 1b–d and Figure 2b–d. We also report the maximum value of m for the DMRG calculations, denoted m_{max}.

For X2C-DMRG calculations of fine-structure splitting, we tested two m values, m = 125 and m = 250, which are considered to be minimum reasonable choices used in practical calculations, providing a fair trade-off between accuracy and computational requirements. For convergence of total energies, we also considered m = 500 to better understand the convergence of electron correlation. For X2C-DMRG calculations of molecular hydrides, we used smaller m values (m = 63 and m = 125) than the atomic systems because the correlation spaces were smaller.

3.1. p-Block Atoms. *3.1.1. ^2P Fine-Structure Splitting.* The ground-state $^2P_{1/2} - {}^2P_{3/2}$ splitting of Ga, In, and Tl was computed with X2C-CASSCF, X2C-MRCISD, and X2C-DMRG (with m = 125 and m = 250), as shown in Table 1. The X2C-SA(6)-CAS(3,8)SCF values are reported because the wave function was used as a reference for both X2C-DMRG and X2C-MRCI. Two different correlation spaces are defined in Figure 1. For correlation spaces **A** and **B**, we find excellent agreement between X2C-DMRG with m = 250 and X2C-MRCISD, whereas X2C-DMRG with m = 125 shows a small deviation with the largest observed error 10 meV relative to X2C-MRCISD for Tl in space **A**. Upon increasing from m = 125 to m = 250, the error of Tl with space **A** decreases to 2 meV.

The X2C-DMRG results in Table 1 showcase its computational advantage because the equivalent CASCI space is not simply feasible using traditional CI methods. For example, the full CAS(27, 64) space for Tl has a total of 3×10^{17} determinants, which is not accessible using even a contemporary MPI-parallelized implementation of the traditional relativistic CASCI approach[80] and computational power.



b)	Ga	
	A(13, 36)	B(3, 58)
N^{MRCISD}	5×10^5	1×10^4
m_{max}	3×10^5	5×10^8

c)	In	
	A(13, 50)	B(3, 72)
N^{MRCISD}	2×10^6	2×10^4
m_{max}	3×10^7	7×10^{10}

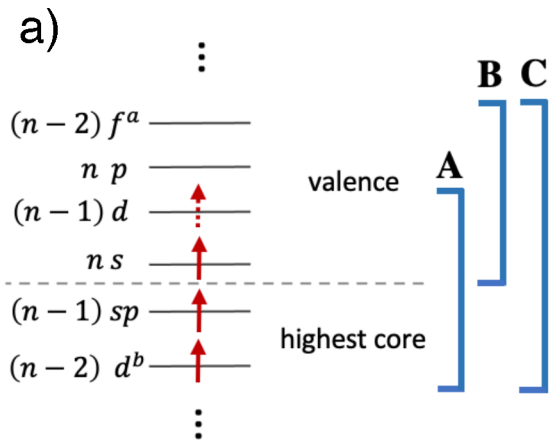
d)	Tl	
	A(27, 64)	B(3, 72)
N^{MRCISD}	9×10^6	2×10^4
m_{max}	4×10^9	7×10^{10}

Figure 1. (a) Definitions of selected correlation spaces. Molecular spinor orbitals are optimized using the X2C-SA(6)-CAS(3,8)SCF procedure. For X2C-MRCI, solid and dotted arrows indicate fully and partially occupied shells, respectively. Note that virtual orbitals are ordered symbolically instead of energetically. (b–d) Total numbers of two-component configurations in X2C-MRCISD and m_{max} for X2C-DMRG.

3.1.2. Correlation Convergence. In this section, we study the convergence profile of the X2C-DMRG method with respect to the m value for the absolute energy of the ground state. We consider two cases of Ga: space **A** and space **B** (see Figure 1). Though both spaces include valence orbitals, space **A** adds subvalence core orbitals and space **B** includes Rydberg virtual orbitals. In Figure 3, we report the difference in ground-state energy relative to X2C-MRCISDT results to gain insight into the degree of electron correlation captured as a function of m. For space **A** with extra core orbitals, reported in Figure 3a, the X2C-DMRG(m=125) ground-state energy is higher than the X2C-MRCISDT result by 25 meV and even higher than X2C-MRCISD by 14 meV. As the m value increases, X2C-DMRG quickly converges the ground state with m = 500 within 1 meV of the reference the X2C-MRCISDT energy. For space **B** with extra Rydberg virtual orbitals, reported in Figure 3b, we find X2C-DMRG with the smallest m value investigated here already better converges the ground state than X2C-

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a: only for La; *b*: only for Y and La

b)	Sc		
	A(11, 20)	B(3, 18)	C(11, 26)
N^{MRCISD}	3×10^4	8×10^2	2×10^5
$m_{\max}$	1×10^3	5×10^2	8×10^3

c)	Y		
	A(21, 30)	B(3, 18)	C(21, 36)
N^{MRCISD}	1×10^5	8×10^2	1×10^6
$m_{\max}$	3×10^4	5×10^2	3×10^5

d)	La		
	A(21, 30)	B(3, 32)	C(21, 50)
N^{MRCISD}	1×10^5	4×10^3	8×10^6
$m_{\max}$	3×10^4	7×10^4	3×10^7

Figure 2. (a) Definitions of selected correlation spaces. Molecular spinor orbitals are optimized using the X2C-SA(10)-CAS(3,12)SCF procedure. For X2C-MRCI, solid and dotted arrows indicate fully and partially occupied shells, respectively. Note that virtual orbitals are ordered symbolically instead of energetically. (b–d) Total numbers of two-component configurations in X2C-MRCISD and $m_{\max}$ for X2C-DMRG.

MRCISD. Again, X2C-DMRG converges faster with virtual orbitals (space **B**) than with core orbitals (space **A**) for the electron correlation.

3.2. d-Block Atoms. *3.2.1. ^2D Fine-Structure Splitting.* The $^2D_{3/2} - {}^2D_{5/2}$ splitting of Sc, Y, and La was computed with X2C-CASSCF, X2C-MRCI, and X2C-DMRG (with m = 125 and m = 250), as shown in Table 2 with correlation space partitioning illustrated in Figure 2. For the **A** and **B** spaces of

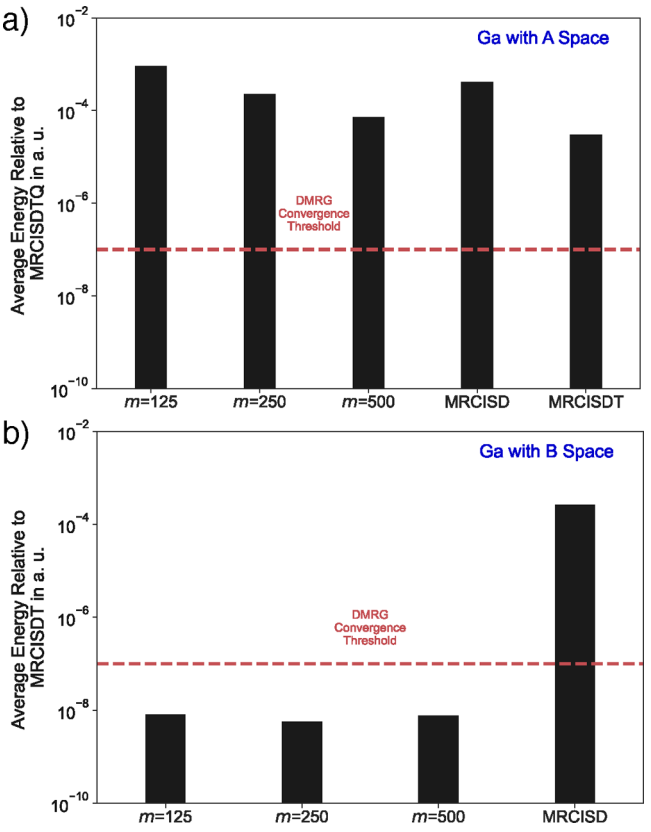


Figure 3. (a) Convergence of electron correlation (relative to X2C-MRCISDT) as a function of m for space **A** of Ga atom. (b) Convergence of electron correlation (relative to X2C-MRCISDT) as a function of m for space **B** of Ga atom. See Figure 1 for the definition of correlation space.

d-block atoms, we find excellent agreement between X2C-DMRG and X2C-MRCISD, where the largest discrepancy is 3 meV for the **A** space of La atom. The X2C-DMRG results for **C** spaces are not converged with respect to m to the same degree as the **A** and **B** spaces for d-block atoms. This is because the increased active space size requires a larger m to reach the full convergence. For Sc, the X2C-DMRG with m = 125 for the **C** space matches the X2C-MRCISD but differs by 2 meV with m = 250. For Y atom, the **C** space differs by 8 meV for m = 125 and 1 meV for m = 250 from X2C-MRCISD. For the La atom, the **C** space is much more challenging than Y atom, which would correspond to an increase from 6×10^9 to 7×10^{13} determinants with traditional CI methods. For both values of m, we find the largest deviations from X2C-MRCISD are 159 and 47 meV for m = 125 and m = 250, respectively. The increase in m from 125 to 250 improves the splitting by 112 meV; however, additional convergence studies with respect to m for La with the **C** space are needed. These

Table 1. $^2P_{1/2} - {}^2P_{3/2}$ Splitting (in meV) for X2C-DMRG (m = 125 and m = 250), X2C-CASSCF, and X2C-MRCISD[a]

		X2C-DMRG(m=125)		X2C-DMRG(m=250)		X2C-MRCISD[32]		
	X2C-CASSCF	A	B	A	B	A	B	expt[79]
Ga	90	89	97	92	97	92	97	102
In	239	239	254	243	254	244	254	274
Tl	863	865	902	859	902	856	903	966

[a]See Figure 1 for the definition of correlation space.

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Table 2. $^2D_{3/2} - {}^2D_{5/2}$ Splitting (in meV) for X2C-DMRG (m = 125 and m = 250), X2C-CASSCF, and X2C-MRCISD[a]

		X2C-DMRG(m=125)			X2C-DMRG(m=250)			X2C-MRCISD			
	X2C-CASSCF	A	B	C	A	B	C	A	B	C	expt[79]
Sc	22	23	22	23	23	22	25	23	22	23	21
Y	55	57	60	57	58	60	64	58	60	65	66
La	92	98	107	275	101	107	163	101	107	116	130

[a]See Figure 2 for the definition of correlation space.

results can be also improved by adding orbitals into the RAS2 space from the RAS1 or RAS3 space or making the correlation space larger. To showcase this, we added six p orbitals to the RAS2 space from RAS3 for the La **B** space, which improved the splitting to 122 meV. Note that difficulty with converging m is not completely explained by the active space size. We find larger errors for the La **C** space than for the Tl **A** space (see Table 1); however, the Tl **A** space is larger than the La **C** space (compare Figures 1 and 2). This agrees with the general consensus that treatment of electron correlation in transition metals is challenging.

We emphasize a practical issue with excited-state calculations with the state-specific DMRG approach. There is no guarantee that the solutions obtained are in energetic-ascending ordering. We found this problematic for the La **C** space, the only system not converged to less than 5 meV with m = 250. We needed to calculate 18 roots for the 10 microstates of the $^2D_{3/2} - {}^2D_{5/2}$ splitting. Because each additional root needs to fully solve the DMRG problem, memory and time requirements are significantly increased. Improved DMRG excited-state treatments for strongly correlated systems with intermediate m values are needed.[73]

3.2.2. Correlation Convergence. To further investigate the convergence profile of the X2C-DMRG method with respect to the m value, we consider spaces **A** and **C** for energy and fine-structure calculations of Sc. The Sc **A** space, CAS(11, 20), includes valence and core orbitals, whereas the Sc **C** space, CAS(11, 26), adds additional unoccupied valence shell orbital. We do not include the **B** space with only a few active electrons because X2C-DMRG converges quickly to the full CI limit. We used X2C-MRCISDTQ56 as the reference in this study.

For the smaller **A** space, the X2C-DMRG total energy with m = 125 already outperforms X2C-MRCISD and X2C-MRCISDT. Figure 4a shows that as m increases, X2C-DMRG quickly converges toward the limit of X2C-MRCISDTQ56. With m = 500, X2C-DMRG agrees best with the X2C-MRCISDTQ56 reference. For the larger **C** space, shown in Figure 4b, the X2C-DMRG energy convergence is slower than in the **A** space. Even with m = 500, the X2C-DMRG energy is still 10^{-4} au higher than the X2C-MRCISDTQ56 reference.

3.3. Symmetry Breaking. The X2C-DMRG method developed here is in the Kramers-unrestricted framework. We found no Kramers' symmetry breaking in the CASSCF reference wave functions in this work (see Supporting Information). In the full CI limit, the Kramers' symmetry of DMRG is fully satisfied. However, in the case of limited m value, it remains to be seen how the variational X2C-DMRG can maintain the Kramers' symmetry. In this section, we investigate the degree of microstate symmetry breaking in the matrix-product-state wave function. The atomic $^2P_{1/2}$ and $^2P_{3/2}$ ($^2D_{3/2}$ and $^2D_{5/2}$) manifolds have a 2- and 4-fold (4- and 6-fold) degeneracy, corresponding to the M_J = ±1/2 and M_J = ±3/2, ±1/2 (M_J = ±3/2, ±1/2 and M_J = ±5/2, ±3/2, ±1/2) microstates, respectively. The ±M_J microstates form a degenerate Kramers' pair, related by the time-reversal symmetry. In the absence of ligand field or external field, all microstates that belong to the same level should be degenerate. In Figure 5a–d, we report the standard deviation (in au) of roots corresponding to the $^2P_{1/2}$, $^2P_{3/2}$, $^2D_{3/2}$ and $^2D_{5/2}$ manifolds for the computed p-block and d-block elements, respectively.

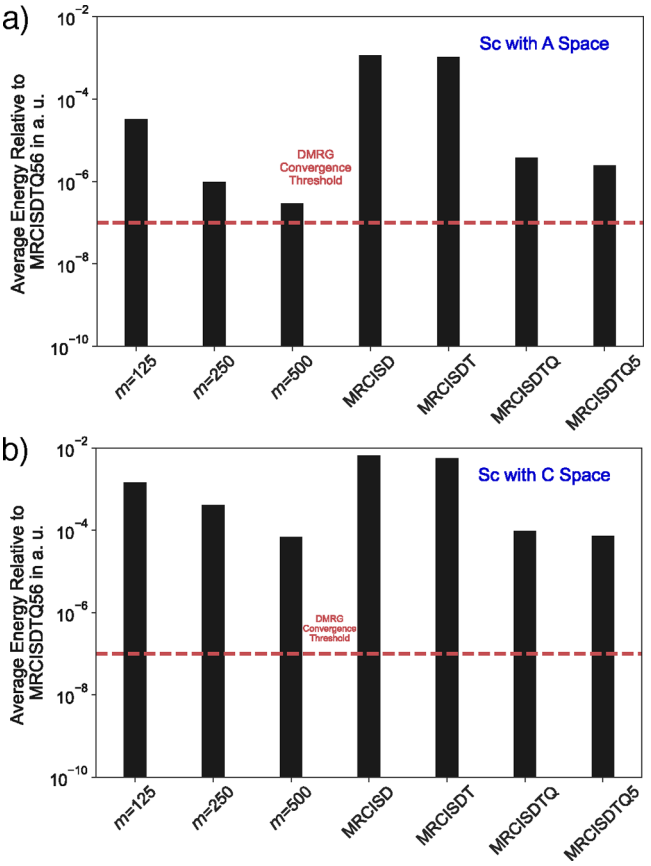


Figure 4. (a) Convergence of electron correlation (relative to MRCISDTQ56) as a function of m for **A** space of Sc atom. (b) Convergence of electron correlation (relative to MRCISDTQ56) as a function of m for **C** space of Sc atom.

Figure 5 shows that all calculations exhibit a certain degree of symmetry breaking, which is strongly dependent on m and the size of the orbital space. We indicate where symmetry breaking is less than 1 meV because we consider this to be the metric for spectroscopic accuracy and showcase where symmetry breaking might affect the reported fine-structure splittings in Tables 1 and 2. In the Supporting Information, we report Kramers-restricted and -unrestricted X2C-MRCISD calculations for the **A** space of Ga atom to further suggest that the symmetry breaking is not because of the use of a Kramers-

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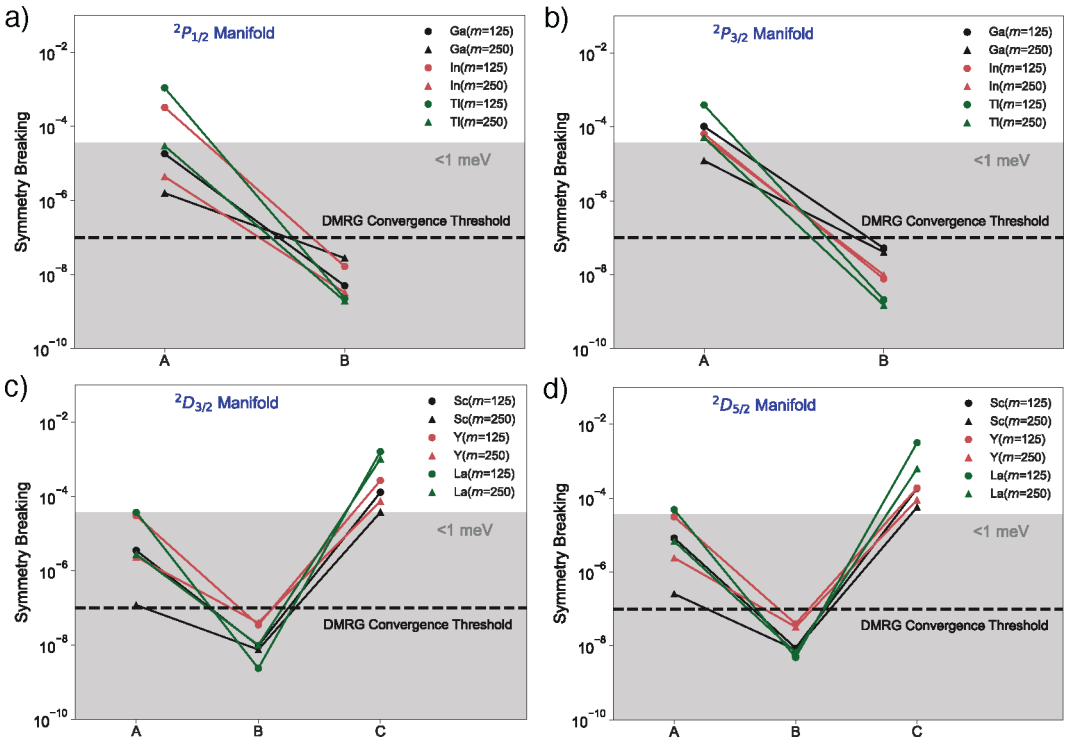


Figure 5. (a,b) Symmetry breaking (in au) of the $^2P_{1/2}$ and $^2P_{3/2}$ manifolds as a function of orbital space for p-block atoms. (c,d) Symmetry breaking (in au) of the $^2D_{3/2}$ and $^2D_{5/2}$ manifolds as a function of orbital space for d-block atoms. The shaded area is defined as the spectroscopic accuracy (<1 meV).

Table 3. Low-Lying Vertical Excitation Energies of TlH (in eV) for X2C-DMRG (m = 63 and m = 125), X2C-CASSCF, X2C-MRCISD, and X2C-MRPT2

	X2C-CASSCF	X2C-DMRG(m=63)	X2C-DMRG(m=125)	X2C-MRCISD	X2C-MRPT2[9]
$^1\Sigma^+_{0^+} \rightarrow {}^3\Pi_{0^-}$	1.719	1.924	1.911	1.906	2.091
$^1\Sigma^+_{0^+} \rightarrow {}^3\Pi_{0^+}$	1.816	2.013	1.999	1.994	2.190
$^1\Sigma^+_{0^+} \rightarrow {}^3\Pi_{1}$	1.936	2.115	2.104	2.100	2.249
$^1\Sigma^+_{0^+} \rightarrow {}^3\Pi_{2}$	2.385	2.538	2.536	2.536	2.730

Table 4. $^2\Pi$ Splitting of GeH and SnH (in eV) for X2C-DMRG (m = 63 and m = 125), X2C-CASSCF, X2C-MRCISD, and X2C-MRPT2

	X2C-CASSCF	X2C-DMRG(m=63)	X2C-DMRG(m=125)	X2C-MRCISD	X2C-MRPT2[9]
GeH	0.115	0.108	0.108	0.107	0.110
SnH	0.276	0.271	0.271	0.271	0.270

unrestricted formalism. Therefore, the symmetry breaking in Kramers-unrestricted DMRG is likely a result of the imbalanced treatment of degenerate electrons along a matrix-product state for finite m. In terms of overall trends, we find that increasing m decreases symmetry breaking. The **A** space of p-block atoms and the **C** space of d-block atoms have a higher degree of symmetry breaking than other cases tested here. Although it may not be surprising because increasing m converges to full CI, we find it encouraging to observe that the symmetry breaking does indeed decrease as m increases. In the case of the **C** space of d-block elements, because of the large active space, the symmetry recovery also requires a large m value. Although the calculations do not reach the spectroscopic accuracy of 1 meV, the trend is readily seen from the analysis.

3.4. Diatomic Hydrides. We investigated the performance of our Kramers-unrestricted X2C-DMRG on molecular systems: the low-lying vertical excitation energies of TlH and the $^2\Pi$ splitting of GeH and SnH. We include X2C-MRPT2 from the literature,[9] where all orbitals were placed in the correlation space, for reference. Because correlating all orbitals for DMRG is not feasible (DMRG for a converged m treats excitations beyond doubles), we recommend comparing to X2C-MRCISD, which uses the same correlation space as X2C-DMRG. Such a comparison allows us to understand how well our X2C-DMRG has converged with respect to m.

For the low-lying excited-state energies of TlH, we consider X2C-MRPT2 to be the more accurate value because the complete correlation space is used, whereas, the X2C-DMRG and X2C-MRCISD results use a correlation space of (14, 30). However, we use X2C-MRCISD as the reference for the X2C-DMRG because they use the same correlation space. In Table 3, we find both X2C-DMRG and X2C-MRCISD agree better

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with the X2C-MRPT2 than X2C-CASSCF, which makes sense because the X2C-CASSCF active space of (4, 10) is much smaller than that of (14, 30). When comparing X2C-DMRG with X2C-MRCISD, we find that m = 63 deviates from the X2C-MRCISD by about 15−20 meV for all excitations except $^1\Sigma^+_{0^+} \rightarrow {}^3\Pi_2$. This is improved to about a 5 meV error for m = 125. This can be further improved by increasing m, where we find that m = 250 improves the $^1\Sigma^+_{0^+} \rightarrow {}^3\Pi_{0^-}$ to a 2 meV error.

In Table 4, we report the $^2\Pi$ splitting of both GeH and SnH. Unlike TlH, we find excellent convergence of X2C-DMRG with respect to X2C-MRCISD at m = 63 (1 meV error), which makes sense because the correlation space is much smaller, (3, 16) compared to (14, 30). Increasing m to 125 does not improve the $^2\Pi$ splitting, indicating that m = 63 is converged to the full CI limit in this space.

4. CONCLUSION

Variational relativistic treatments of electronic structure are useful tools for accurately including spin−orbit coupling for relativistic spectroscopies and heavy-element chemistry. However, such applications are often multireference, where more than one configuration is needed for qualitative accuracy. Traditional multireference methods are bottlenecked by the large configurational space in the configuration interaction (CI) framework, where $\sim 10^{12}$ determinants are often considered the limit in the field. We present a Kramers-unrestricted variational two-component density matrix renormalization group (X2C-DMRG), capable of going beyond the limit of traditional CI methods for systems with non-negligible spin−orbit coupling.

We benchmarked against fine-structure splittings of p- and d-block atoms and excitation energies of molecular hydrides. We show that (a) relativistic active spaces resulting in up to 10^{17} configurations are doable with X2C-DMRG, (b) atomic fine-structure splittings are reproduced for most cases relative to X2C-MRCISD, (c) electron correlation is efficiently recovered, yielding results of X2C-MRCISDTQ5 quality in some cases, (d) symmetry breaking within the atomic multiplet is present with X2C-DMRG, which can result in standard deviations up to 30 meV, (e) the origin of the symmetry breaking in X2C-DMRG is likely because of unconverged m as opposed to the Kramers-unrestricted formalism, and (f) excitation energies of monohydrides exhibit similar convergence profiles with respect to m as the atomic fine-structure splittings. We conclude that X2C-DMRG is an efficient tool for approaching the full configuration interaction limit with a relativistic Hamiltonian.

APPENDIX

Explicit Treatment of Spin−Orbit Coupling

In this section, we briefly review the relativistic electronic Hamiltonian and two-component formalism. In the spinor basis, the matrix representation of the restricted kinetically-balanced Dirac equation (in au) is[18,19,81,82]

$$\begin{pmatrix} \mathbf{V} & \mathbf{T} \\ \mathbf{T} & \frac{1}{4c^2}\mathbf{W} - \mathbf{T} \end{pmatrix} \begin{pmatrix} \mathbf{C}_\mathrm{L}^+ & \mathbf{C}_\mathrm{L}^- \\ \mathbf{C}_\mathrm{S}^+ & \mathbf{C}_\mathrm{S}^- \end{pmatrix} = \begin{pmatrix} \mathbf{S} & \mathbf{0}_2 \\ \mathbf{0}_2 & \frac{1}{2c^2}\mathbf{T} \end{pmatrix} \begin{pmatrix} \mathbf{C}_\mathrm{L}^+ & \mathbf{C}_\mathrm{L}^- \\ \mathbf{C}_\mathrm{S}^+ & \mathbf{C}_\mathrm{S}^- \end{pmatrix} \begin{pmatrix} \epsilon^+ & \mathbf{0} \\ \mathbf{0} & \epsilon^- \end{pmatrix} \tag{4}$$

where c is the speed of light. $\mathbf{V}$, $\mathbf{T}$, and $\mathbf{S}$ are the two-component nonrelativistic potential energy, kinetic energy, and overlap matrices, respectively. In eq 4, $\mathbf{W}$ is the relativistic potential matrix, defined as $(\vec{\sigma}\cdot\vec{p})\mathbf{V}(\vec{\sigma}\cdot\vec{p})$, where $\vec{p}$ is the linear momentum operator and $\vec{\sigma}$ contains Pauli spin matrices. The sets of positive/negative eigenvalues, $\{\epsilon^+\}$, $\{\epsilon^-\}$, correspond to the molecular orbital coefficients $(\mathbf{C}_\mathrm{L}^+ \quad \mathbf{C}_\mathrm{S}^+)^\mathrm{T}$ for the positive- and $(\mathbf{C}_\mathrm{L}^- \quad \mathbf{C}_\mathrm{S}^-)^T$ for the negative-energy solutions.

The negative-energy solutions are often not of interest in electronic structure problems. We make use of the 1-electron exact-two-component (X2C) approach to fold in the negative-energy part and work directly with the positive-energy spectrum, which correspond to electronic solutions.[32,43−61] The X2C transformation yields a set of spinor molecular orbitals, which are complex linear combinations of AO spin−orbitals. Spinor molecular orbitals (MOs) can be written as

$$\psi_i(\mathbf{x}) = \begin{pmatrix} \phi_i^\alpha(\mathbf{r}) \\ \phi_i^\beta(\mathbf{r}) \end{pmatrix} \tag{5}$$

where the spatial functions $\{\varphi_k^\alpha(\mathbf{r})\}$and $\{\varphi_k^\beta(\mathbf{r})\}$ are expanded in terms of a common set of real atomic orbital basis functions with complex-valued MO coefficients, $\{c_{\mu i}^\alpha\}$ and $\{c_{\mu i}^\beta\}$. We use one-electron X2C coupled with the Boettger scaling factor (to empirically account for neglect of two-electron spin−orbit coupling) for including relativistic effects.[42]

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jpca.2c02150.

Weights (CI coefficient squared) of the dominant configuration in the CASSCF reference wave functions for atomic multiplet splittings, X2C-CASSCF and X2C-DMRG total energies for atomic systems, and comparison between Kramers-restricted and Kramers-unrestricted X2C-MRCISD for $^2P_{1/2} - {}^2P_{3/2}$ splitting of Ga atom with the **A** space (PDF)

AUTHOR INFORMATION

Corresponding Authors

Stefan Knecht − *Algorithmiq Ltd., FI-00160 Helsinki, Finland; Abteilung SHE Chemie, GSI Helmholtzzentrum für Schwerionenforschung, DE-64291 Darmstadt, Germany; Department Chemie, Johannes-Gutenberg Universität Mainz, DE-55128 Mainz, Germany*; Email: stefan@algorithmiq.fi

Xiaosong Li − *Department of Chemistry, University of Washington, Seattle, Washington 98195, United States*; orcid.org/0000-0001-7341-6240; Email: xsli@uw.edu

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Authors

Chad E. Hoyer − *Department of Chemistry, University of Washington, Seattle, Washington 98195, United States*

Hang Hu − *Department of Chemistry, University of Washington, Seattle, Washington 98195, United States*

Lixin Lu − *Department of Chemistry, University of Washington, Seattle, Washington 98195, United States;* orcid.org/0000-0003-3203-4822

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.jpca.2c02150

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Fedorov, D. G.; Koseki, S.; Schmidt, M. W.; Gordon, M. S. Spin-orbit Coupling in Molecules: Chemistry Beyond the Adiabatic Approximation. *Int. Rev. Phys. Chem.* **2003**, *22*, 551−592.

(2) Pyykkö, P. Relativistic Effects in Chemistry: More Common Than You Thought. *Annu. Rev. Phys. Chem.* **2012**, *63*, 45−64.

(3) Marian, C. M. Spin-Orbit Coupling and Intersystem Crossing in Molecules. *WIREs Comp. Mol. Sci.* **2012**, *2*, 187−203.

(4) Valentine, A. J. S.; Li, X. Intersystem Crossings in Late-Row Elements: A Perspective. *J. Phys. Chem. Lett.* **2022**, *13*, 3039−3046.

(5) Schmidt, M. W.; Gordon, M. S. The Construction and Interpretation of MCSCF Wavefunctions. *Annu. Rev. Phys. Chem.* **1998**, *49*, 233−266.

(6) Roos, B. O.; Taylor, P. R.; Sigbahn, P. E.M. A Complete Active Space SCF Method (CASSCF) Using a Density Matrix Formulated Super-CI Approach. *Chem. Phys.* **1980**, *48*, 157−173.

(7) Andersson, K.; Malmqvist, P.-Å.; Roos, B. O.; Sadlej, A. J.; Wolinski, K. Second-Order Perturbation Theory with a CASSCF Reference Function. *J. Phys. Chem.* **1990**, *94*, 5483−5488.

(8) Andersson, K.; Malmqvist, P.-Å.; Roos, B. O. Second-Order Perturbation Theory with a Complete Active Space Self Consistent Field Reference Function. *J. Chem. Phys.* **1992**, *96*, 1218−1226.

(9) Lu, L.; Hu, H.; Jenkins, A. J.; Li, X. Exact-Two-Component Relativistic Multireference Second-Order Perturbation Theory. *J. Chem. Theory Comput.* **2022**, *18*, 2983−2992.

(10) Aquilante, F.; Autschbach, J.; Carlson, R. K.; Chibotaru, L. F.; Delcey, M. G.; De Vico, L.; Ferré, N.; Frutos, L. M.; Gagliardi, L.; Garavelli, M.; et al. Molcas 8: New Capabilities for Multiconfigurational Quantum Chemical Calculations Across the Periodic Table. *J. Comput. Chem.* **2016**, *37*, 506−541.

(11) White, S. R. Density Matrix Formulation for Quantum Renormalization Groups. *Phys. Rev. Lett.* **1992**, *69*, 2863−2866.

(12) Schollwöck, U. The Density-Matrix Renormalization Group in the Age of Matrix Product States. *Ann. Phys.* **2011**, *326*, 96−192.

(13) Chan, G. K.-L.; Sharma, S. The Density Matrix Renormalization Group in Quantum Chemistry. *Annu. Rev. Phys. Chem.* **2011**, *62*, 465−481.

(14) Wouters, S.; Van Neck, D. The Density Matrix Renormalization Group for Ab Initio Quantum Chemistry. *Euro. Phys. J. D* **2014**, *68*, 272.

(15) Baiardi, A.; Reiher, M. The Density Matrix Renormalization Group in Chemistry and Molecular Physics: Recent Developments and New Challenges. *J. Chem. Phys.* **2020**, *152*, 040903.

(16) Roos, B. O.; Malmqvist, P.-Å. Relativistic Quantum Chemistry: The Multiconfigurational Approach. *Phys. Chem. Chem. Phys.* **2004**, *6*, 2919−2927.

(17) Cao, Z.; Li, Z.; Wang, F.; Liu, W. Combining the Spin-separated Exact Two-component Relativistic Hamiltonian with the Equation-of-Motion Coupled-Cluster Method for the Treatment of Spin-orbit Splittings of Light and Heavy Elements. *Phys. Chem. Chem. Phys.* **2017**, *19*, 3713−3721.

(18) Dyall, K. G.; Fægri, K., Jr. *Introduction to Relativistic Quantum Chemistry*; Oxford University Press: New York, 2007.

(19) Reiher, M.; Wolf, A. *Relativistic Quantum Chemistry*, 2nd ed.; Wiley-VCH: Weinheim, 2015.

(20) Rota, J.-B.; Knecht, S.; Fleig, T.; Ganyushin, D.; Saue, T.; Neese, F.; Bolvin, H. Zero Field Splitting of the Chalcogen Diatomics using Relativistic Correlated Wave-Function Methods. *Theor. Chem. Acc.* **2011**, *135*, 114106.

(21) Knecht, S.; Jensen, H. J. A.; Saue, T. Relativistic Quantum Chemical Calculations Show that the Uranium Molecule U_2 has a Quadruple Bond. *Nat. Chem.* **2019**, *11*, 40−44.

(22) Jensen, H. J. A.; Dyall, K. G.; Saue, T.; Fægri, K. Relativistic Four-component Multiconfigurational Self-Consistent-Field Theory for Molecules: Formalism. *J. Chem. Phys.* **1996**, *104*, 4083−4097.

(23) Thyssen, J.; Fleig, T.; Jensen, H. J. A. A Direct Relativistic Four-component Multiconfiguration Self-Consistent-Field Method for Molecules. *J. Chem. Phys.* **2008**, *129*, 034109.

(24) Almoukhalalati, A.; Knecht, S.; Jensen, H. J. A.; Dyall, K. G.; Saue, T. Electron Correlation within the Relativistic No-pair Approximation. *J. Chem. Phys.* **2016**, *145*, 074104.

(25) Bates, J. E.; Shiozaki, T. Fully Relativistic Complete Active Space Self-Consistent Field for Large Molecules: Quasi-second-order Minimax Optimization. *J. Chem. Phys.* **2015**, *142*, 044112.

(26) Shiozaki, T.; Mizukami, W. Relativistic Internally Contracted Multireference Electron Correlation Methods. *J. Chem. Theory Comput.* **2015**, *11*, 4733−4739.

(27) Vlaisavljevich, B.; Shiozaki, T. Nuclear Energy Gradients for Internally Contracted Complex Active Space Second-Order Perturbation Theory: Multistate Extensions. *J. Chem. Theory Comput.* **2016**, *12*, 3781−3787.

(28) Reynolds, R. D.; Yanai, T.; Shiozaki, T. Large-scale Relativistic Complete Active Space Self-Consistent Field with Robust Convergence. *J. Chem. Phys.* **2018**, *149*, 014106.

(29) Knecht, S.; Legeza, Ö.; Reiher, M. Communication: Four-Component Density Matrix Renormalization Group. *J. Chem. Phys.* **2014**, *140*, 041101.

(30) Battaglia, S.; Keller, S.; Knecht, S. Efficient Relativistic Density-Matrix Renormalization Group Implementation in a Matrix-Product Formulation. *J. Chem. Theory Comput.* **2018**, *14*, 2353−2369.

(31) Jenkins, A. J.; Liu, H.; Kasper, J. M.; Frisch, M. J.; Li, X. Variational Relativistic Complete Active Space Self-Consistent Field Method. *J. Chem. Theory Comput.* **2019**, *15*, 2974−2982.

(32) Hu, H.; Jenkins, A. J.; Liu, H.; Kasper, J. M.; Frisch, M. J.; Li, X. Relativistic Two-Component Multireference Configuration Interaction Method with Tunable Correlation Space. *J. Chem. Theory Comput.* **2020**, *16*, 2975−2984.

(33) Jenkins, A. J.; Hu, H.; Lu, L.; Frisch, M. J.; Li, X. Two-Component Multireference Restricted Active Space Configuration Interaction for the Computation of L-Edge X-ray Absorption Spectra. *J. Chem. Theory Comput.* **2022**, *18*, 141−150.

(34) Sharma, P.; Jenkins, A. J.; Scalmani, G.; Frisch, M. J.; Truhlar, D. G.; Gagliardi, L.; Li, X. Exact-Two-Component Multiconfiguration Pair-Density Functional Theory. *J. Chem. Theory Comput.* **2022**, *18*, 2947−2954.

(35) Grofe, A.; Li, X. Relativistic Nonorthogonal Configuration Interaction: Application to $L_{2,3}$-edge X-ray Spectroscopy. *Phys. Chem. Chem. Phys.* **2022**, *24*, 10745−10756.

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(36) Sun, S.; Williams-Young, D.; Stetina, T. F.; Li, X. Generalized Hartree-Fock with Non-perturbative Treatment of Strong Magnetic Field: Application to Molecular Spin Phase Transition. *J. Chem. Theory Comput.* **2019**, *15*, 348−356.
(37) Sun, S.; Williams-Young, D.; Li, X. An Ab Initio Linear Response Method for Computing Magnetic Circular Dichroism Spectra with Nonperturbative Treatment of Magnetic Field. *J. Chem. Theory Comput.* **2019**, *15*, 3162−3169.
(38) Stetina, T. F.; Sun, S.; Williams-Young, D. B.; Li, X. Modeling Magneto-Photoabsorption Using Time-Dependent Complex Generalized Hartree-Fock. *ChemPhotoChem.* **2019**, *3* (9), 739−746.
(39) Sun, S.; Beck, R.; Williams-Young, D. B.; Li, X. Simulating Magnetic Circular Dichroism Spectra with Real-Time Time-Dependent Density Functional Theory in Gauge Including Atomic Orbitals. *J. Chem. Theory Comput.* **2019**, *15*, 6824−6831.
(40) Sun, S.; Li, X. Relativistic Effects in Magnetic Circular Dichroism: Restricted Magnetic Balance and Temperature Dependence. *J. Chem. Theory Comput.* **2020**, *16*, 4533−4542.
(41) Kasper, J. M.; Jenkins, A. J.; Sun, S.; Li, X. Perspective on Kramers Symmetry Breaking and Restoration in Relativistic Electronic Structure Methods for Open-Shell Systems. *J. Chem. Phys.* **2020**, *153*, 090903.
(42) Boettger, J. C. Approximate Two-Electron Spin-Orbit Coupling Term For Density-Functional-Theory DFT Calculations Using The Douglas-Kroll-Hess Transformation. *Phys. Rev. B* **2000**, *62*, 7809−7815.
(43) Kutzelnigg, W.; Liu, W. Quasirelativistic Theory Equivalent to Fully Relativistic Theory. *J. Chem. Phys.* **2005**, *123*, 241102.
(44) Liu, W.; Peng, D. Infinite-Order Quasirelativistic Density Functional Method Based on the Exact Matrix Quasirelativistic Theory. *J. Chem. Phys.* **2006**, *125*, 044102.
(45) Peng, D.; Liu, W.; Xiao, Y.; Cheng, L. Making Four- and Two-component Relativistic Density Functional Methods fully Equivalent based on the Idea of 'from Atoms to Molecule. *J. Chem. Phys.* **2007**, *127*, 104106.
(46) Ilias, M.; Saue, T. An Infinite-Order Relativistic Hamiltonian by a Simple One-Step Transformation. *J. Chem. Phys.* **2007**, *126*, 064102.
(47) Liu, W.; Peng, D. Exact Two-component Hamiltonians Revisited. *J. Chem. Phys.* **2009**, *131*, 031104.
(48) Liu, W. Ideas of Relativistic Quantum Chemistry. *Mol. Phys.* **2010**, *108*, 1679−1706.
(49) Saue, T. Relativistic Hamiltonians for Chemistry: A Primer. *ChemPhysChem* **2011**, *12*, 3077−3094.
(50) Li, Z.; Xiao, Y.; Liu, W. On the Spin Separation of Algebraic Two-Component Relativistic Hamiltonians. *J. Chem. Phys.* **2012**, *137*, 154114.
(51) Peng, D.; Middendorf, N.; Weigend, F.; Reiher, M. An Efficient Implementation of Two-Component Relativistic Exact-Decoupling Methods for Large Molecules. *J. Chem. Phys.* **2013**, *138*, 184105.
(52) Egidi, F.; Goings, J. J.; Frisch, M. J.; Li, X. Direct Atomic-Orbital-Based Relativistic Two-Component Linear Response Method for Calculating Excited-State Fine Structures. *J. Chem. Theory Comput.* **2016**, *12*, 3711−3718.
(53) Goings, J. J.; Kasper, J. M.; Egidi, F.; Sun, S.; Li, X. Real Time Propagation of the Exact Two Component Time-Dependent Density Functional Theory. *J. Chem. Phys.* **2016**, *145*, 104107.
(54) Konecny, L.; Kadek, M.; Komorovsky, S.; Malkina, O. L.; Ruud, K.; Repisky, M. Acceleration of Relativistic Electron Dynamics by Means of X2C Transformation: Application to the Calculation of Nonlinear Optical Properties. *J. Chem. Theory Comput.* **2016**, *12*, 5823−5833.
(55) Egidi, F.; Sun, S.; Goings, J. J.; Scalmani, G.; Frisch, M. J.; Li, X. Two-Component Non-Collinear Time-Dependent Spin Density Functional Theory for Excited State Calculations. *J. Chem. Theory Comput.* **2017**, *13*, 2591−2603.
(56) Petrone, A.; Williams-Young, D. B.; Sun, S.; Stetina, T. F.; Li, X. An Efficient Implementation of Two-Component Relativistic Density Functional Theory with Torque-Free Auxiliary Variables. *Euro. Phys. J. B* **2018**, *91*, 169.
(57) Liu, W.; Xiao, Y. Relativistic Time-dependent Density Functional Theories. *Chem. Soc. Rev.* **2018**, *47*, 4481−4509.
(58) Hoyer, C. E.; Williams-Young, D. B.; Huang, C.; Li, X. Embedding Non-Collinear Two-Component Electronic Structure in a Collinear Quantum Environment. *J. Chem. Phys.* **2019**, *150*, 174114.
(59) Stetina, T. F.; Kasper, J. M.; Li, X. Modeling $L_{2/3}$-Edge X-ray Absorption Spectroscopy with Linear Response Exact Two-Component Relativistic Time-Dependent Density Functional Theory. *J. Chem. Phys.* **2019**, *150*, 234103.
(60) Zhang, T.; Kasper, J. M.; Li, X. Localized Relativistic Two-Component Methods for Ground and Excited State Calculations. In *Annual Reports in Computational Chemistry*; Dixon, D. A., Ed.; Elsevier, 2020; Vol. *16*, Chapter 2, pp 17−37.
(61) Hoyer, C. E.; Li, X. Relativistic Two-Component Projection-Based Quantum Embedding for Open-Shell Systems. *J. Chem. Phys.* **2020**, *153*, 094113.
(62) Grofe, A.; Gao, J.; Li, X. Exact-Two-Component Block-Localized Wave Function: A Simple Scheme for the Automatic Computation of Relativistic ΔSCF. *J. Chem. Phys.* **2021**, *155*, 014103.
(63) White, S. R. Density-Matrix Algorithms for Quantum Renormalization Groups. *Phys. Rev. B* **1993**, *48*, 10345.
(64) Marti, K. H.; Ondík, I. M.; Moritz, G.; Reiher, M. Density Matrix Renormalization Group Calculations on Relative Energies of Transition Metal Complexes and Clusters. *J. Chem. Phys.* **2008**, *128*, 014104.
(65) Kurashige, Y.; Yanai, T. Second-Order Perturbation Theory with a Density Matrix Renormalization Group Self-Consistent Field Reference Function: Theory and Application to the Study of Chromium Dimer. *J. Chem. Phys.* **2011**, *135*, 094104.
(66) Sharma, S.; Chan, G. K.-L. Spin-Adapted Density Matrix Renormalization Group Algorithms for Quantum Chemistry. *J. Chem. Phys.* **2012**, *136*, 124121.
(67) Verstraete, F.; Porras, D.; Cirac, J. Density Matrix Renormalization Group and Periodic Boundary Conditions: A Quantum Information Perspective. *Phys. Rev. Lett.* **2004**, *93*, 227205.
(68) McCulloch, I. P. From Density-Matrix Renormalization Group to Matrix Product States. *J. Stat. Mech.* **2007**, *2007*, P10014.
(69) Dorando, J. J.; Hachmann, J.; Chan, G. K.-L. Targeted Excited State Algorithms. *J. Chem. Phys.* **2007**, *127*, 084109.
(70) Hu, W.; Chan, G. K.-L. Excited-State Geometry Optimization with the Density Matrix Renormalization Group, as Applied to Polyenes. *J. Chem. Theory Comput.* **2015**, *11*, 3000−3009.
(71) Ma, Y.; Knecht, S.; Reiher, M. Multiconfigurational Effects in Theoretical Resonance Raman Spectra. *ChemPhysChem* **2017**, *18*, 384−393.
(72) Baiardi, A.; Stein, C. J.; Barone, V.; Reiher, M. Optimization of Highly Excited Matrix Product States with an Application to Vibrational Spectroscopy. *J. Chem. Phys.* **2019**, *150*, 094113.
(73) Baiardi, A.; Kelemen, A. K.; Reiher, M. Excited-State DMRG Made Simple with FEAST. *J. Chem. Theory Comput.* **2022**, *18*, 415−430.
(74) Keller, S.; Dolfi, M.; Troyer, M.; Reiher, M. An Efficient Matrix Product Operator Representation of the Quantum Chemical Hamiltonian. *J. Chem. Phys.* **2015**, *143*, 244118.
(75) Keller, S.; Reiher, M. Spin-Adapted Matrix Product States and Operators. *J. Chem. Phys.* **2016**, *144*, 134101.
(76) Knecht, S.; Hedegaard, E. D.; Keller, S.; Kovyrshin, A.; Ma, Y.; Muolo, A.; Stein, C. J.; Reiher, M. New Approaches for Ab Initio Calculations of Molecules with Strong Electron Correlation. *Chimia* **2016**, *70*, 244−251.
(77) Williams-Young, D. B.; Petrone, A.; Sun, S.; Stetina, T. F.; Lestrange, P.; Hoyer, C. E.; Nascimento, D. R.; Koulias, L.; Wildman, A.; Kasper, J.; Goings, J. J.; Ding, F.; DePrince, A. E., III; Valeev, E. F.; Li, X. The Chronus Quantum (ChronusQ) Software Package. *WIREs Comput. Mol. Sci.* **2020**, *10*, No. e1436.
(78) Pollak, P.; Weigend, F. Segmented Contracted Error-Consistent Basis Sets of Double- and Triple-Valence Quality for One- and Two-Component Relativistic All-Electron Calculations. *J. Chem. Theory Comput.* **2017**, *13*, 3696−3705.

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https://doi.org/10.1021/acs.jpca.2c02150
J. Phys. Chem. A 2022, 126, 5011−5020

(79) Kramida, A.; Ralchenko, Y.; Reader, J.; NIST ASD Team. *NIST Atomic Spectra Database* (version 5.6.1); National Institute of Standards and Technology: Gaithersburg, MD, 2018; https://physics.nist.gov/asd.
(80) Knecht, S.; Jensen, H. J. A.; Fleig, T. Large-scale Parallel Configuration Interaction. II. Two and Four-Component Double-Group General Active Space Implementation with Application to BiH. *J. Chem. Phys.* **2010**, *132*, 014108.
(81) Sun, S.; Stetina, T. F.; Zhang, T.; Hu, H.; Valeev, E. F.; Sun, Q.; Li, X. Efficient Four-Component Dirac−Coulomb−Gaunt Hartree−Fock in the Pauli Spinor Representation. *J. Chem. Theory Comput.* **2021**, *17*, 3388−3402.
(82) Sun, S.; Ehrman, J. N.; Sun, Q.; Li, X. Efficient Evaluation of the Breit Operator in the Pauli Spinor Basis. *J. Chem. Phys.* **2022**, DOI: 10.1063/5.0098828.

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